(a) Program code:

```
clear;
clc;
xdel(winsid()); //close all windows
s=%s/2/%pi;
G=syslin('c',20*(s^2+s+0.5),s*(s+1)*(s+10));
disp(G,'The given transfer function G(s)=')
a=gca();
a.clip_state='on';
nyquist(G,-1000,1000);
xgrid(color('gray'));
a.data_bounds=[-1 -3;3 3];
a.box='on';
```

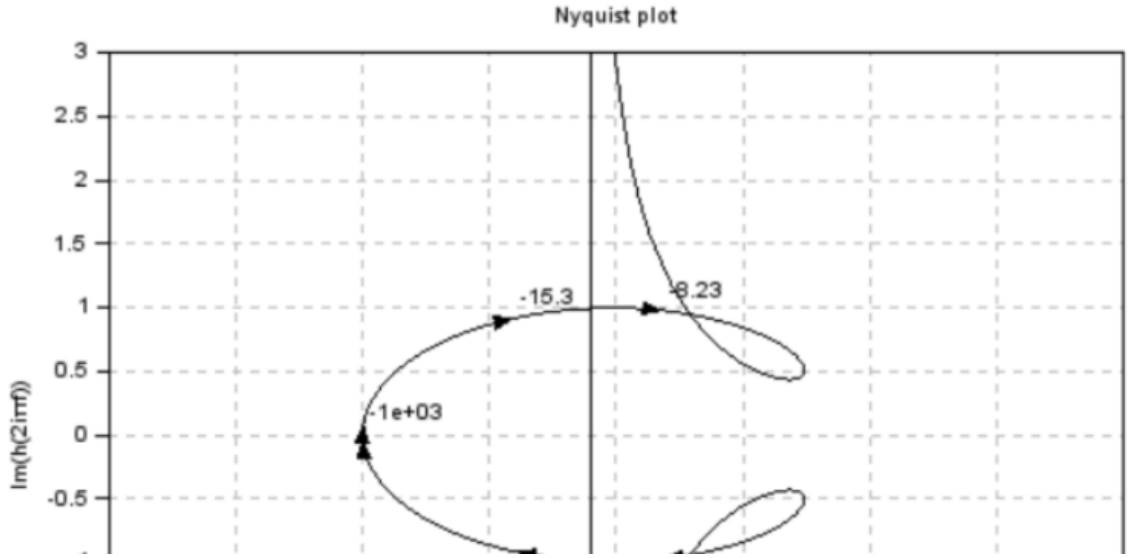
OUTPUT:

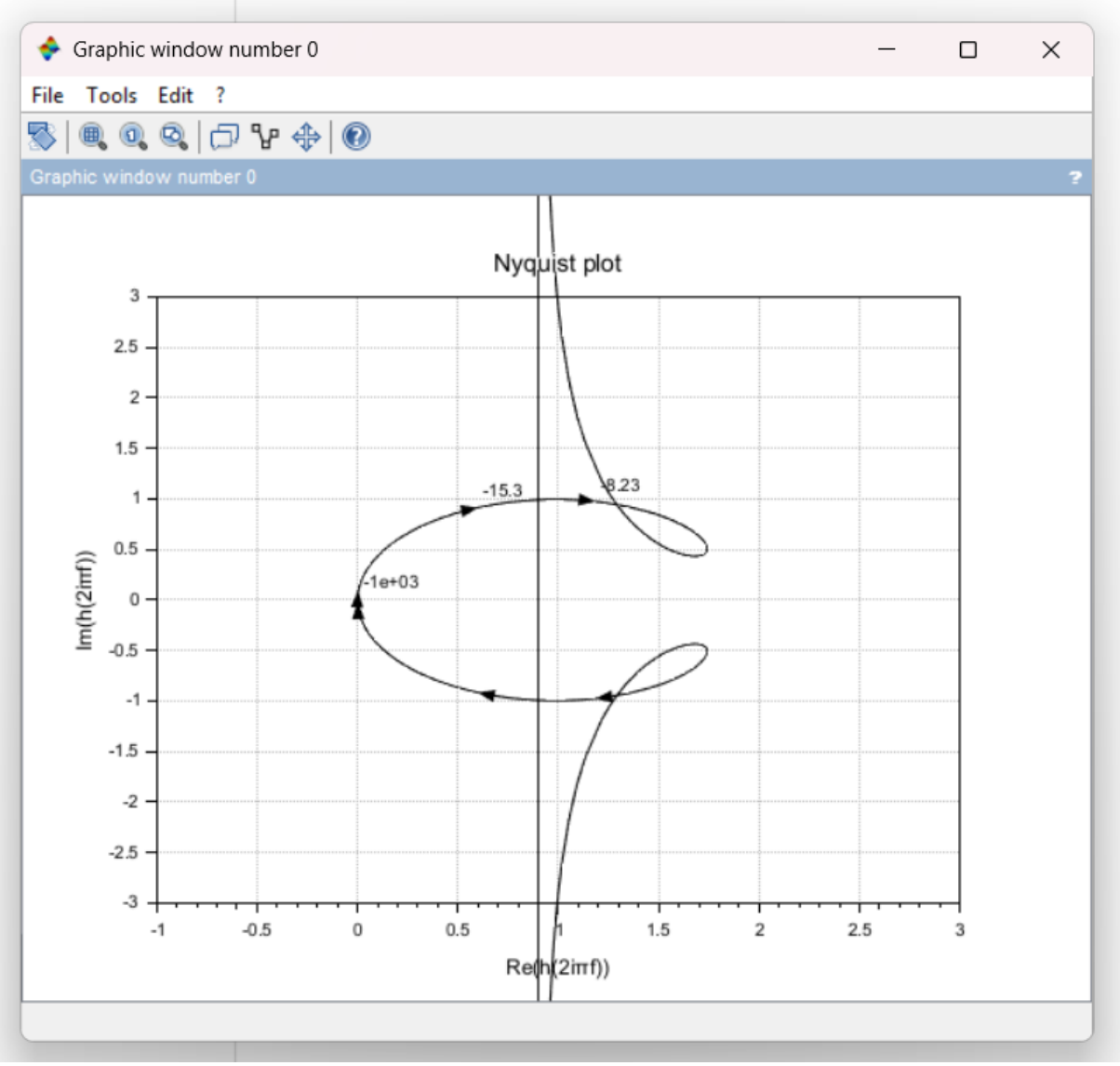
```
The given transfer function G(s)=

                  2
   10 + 20s + 20s
   --------------
            2    3
   10s + 11s + s

-->
```

Graphic window number 0






```
clear;
clc;
xdel(winsid()); //close all windows
s=%s;
T=syslin('c',25+30*s+5*s^2,168+206*s+89*s^2+16*s^3+s^4);
disp(T, 'The given loop transfer function G(s)H(s) = ');
nyquist(T)
a=gca();
a.clip_state='on';
```

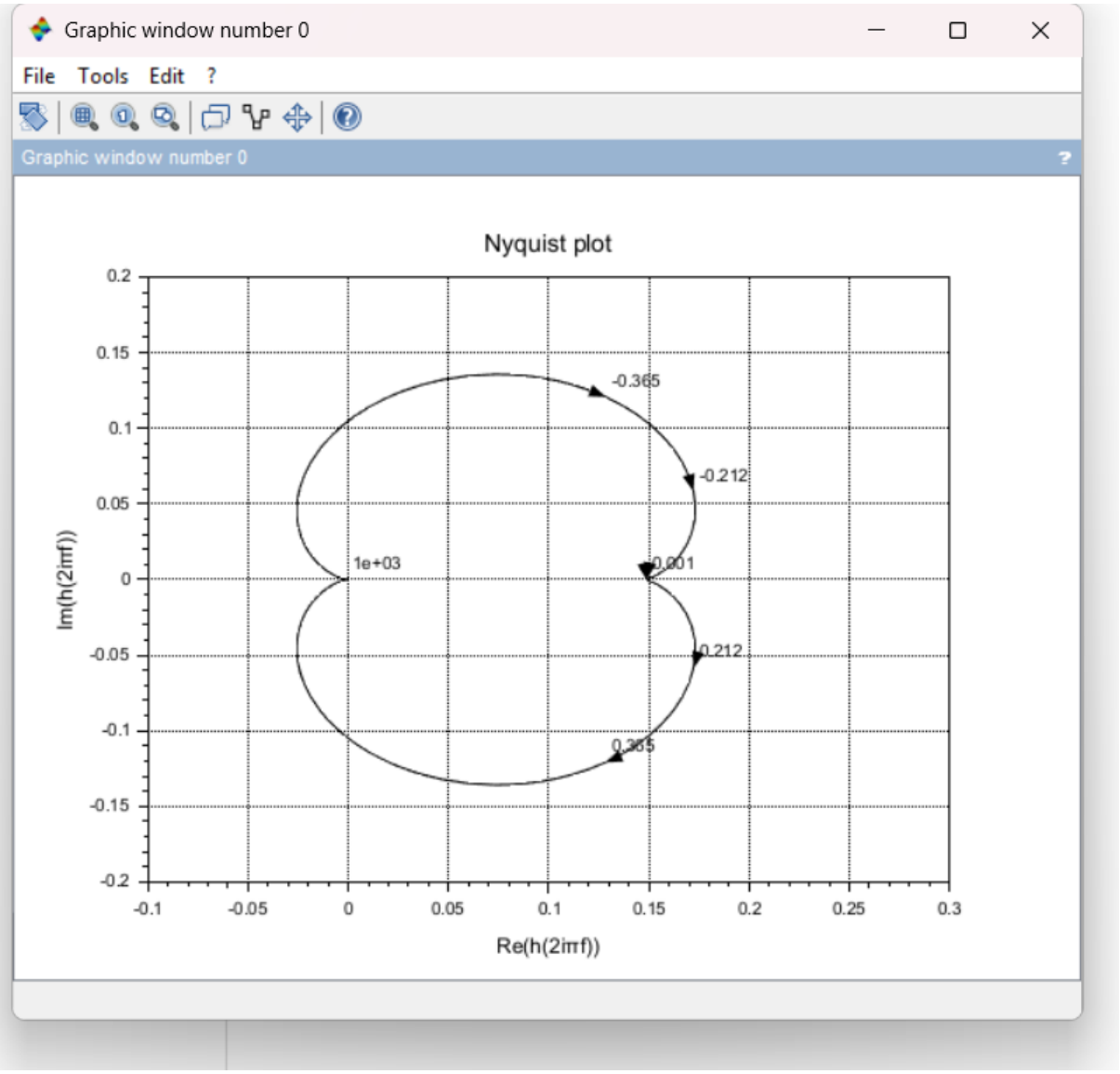
a.data_bounds=[-0.1 -0.2;0.3 0.2];

a.box='on';

Simulation output:

```
Scilab 2025.1.0 Console
          25 +30s +5s^2
   ----------------------------
   168 +206s +89s^2 +16s^3 +s^4
  "The given loop transfer function G(s)H(s) = "

-->
```

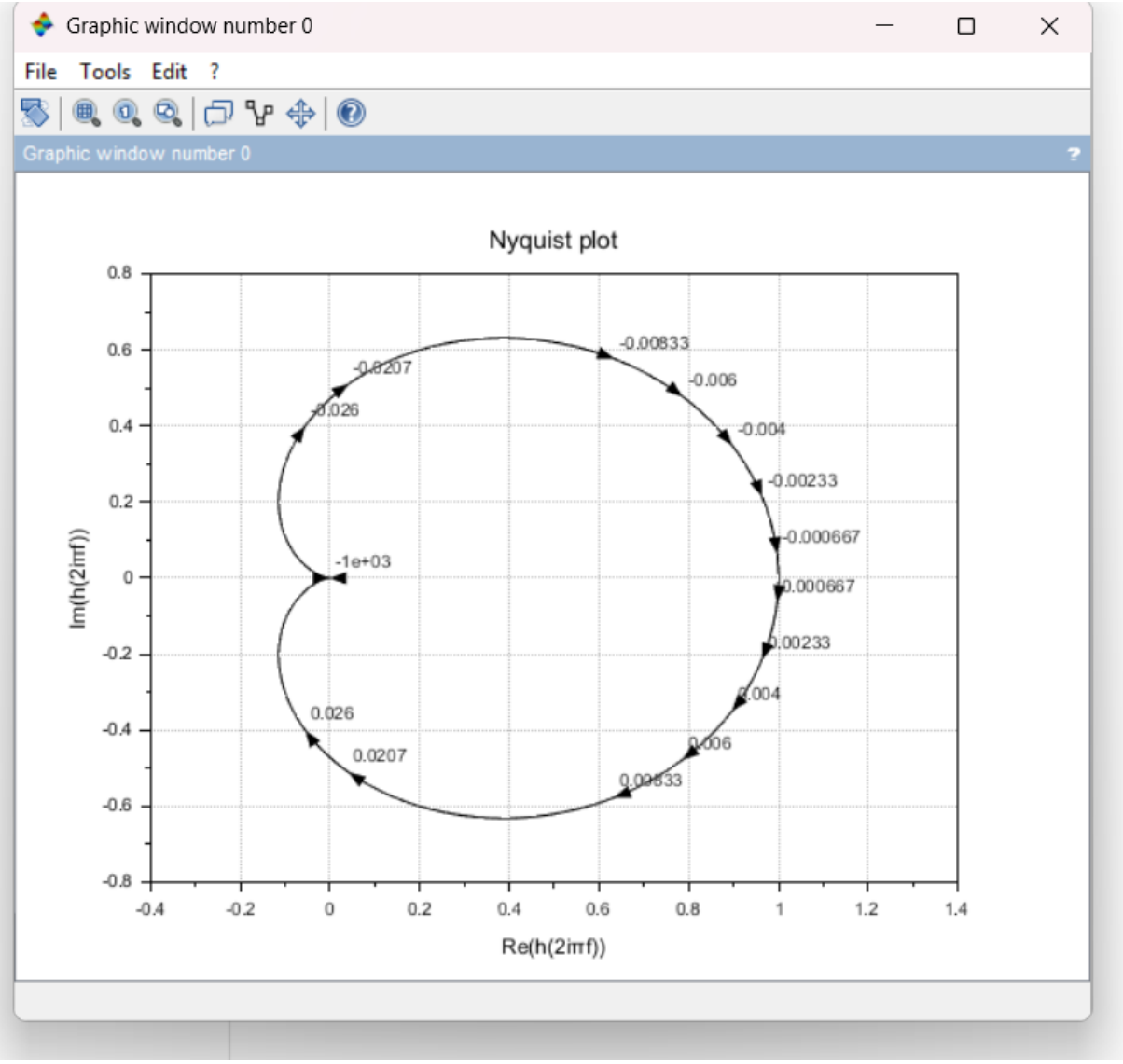


(c) Program code:

```
clear;
clc;
xdel(winsid()); //close all windows
s=%s;
T1=5; T2=10;
K=1;
den=(T1*s+1)*(T2*s+1);
GH=syslin('c',K,den); disp(GH, 'The given transfer function G(s)H(s)=')
nyquist(GH,-1000,1000);
xgrid(color('gray'));
a=gca();
a.clip_state='on';
a.data_bounds=[-0.3 -0.8;1.3 0.8];
a.box='on';
```

OUTPUT:




Graphic window number 0
File Tools Edit ?
Graphic window number 0
Nyquist plot
-0.00833
-0.006
-0.004
-0.00233
-0.000667
0.000667
0.00233
0.004
0.006
0.00833
0.0207
0.026
-1e+03
-0.0207
-0.026
Im(h(2iπf))
Re(h(2iπf))
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-0.4
-0.2
0
0.2
0.4
0.6
0.8
1
1.2
1.4